

Tableau Training – Beginners

A comprehensive guide to data visualization with Tableau Public

Module 1: The World of Data

Understanding the "Why" behind data visualization.

What is Data Visualization?

What is Data Visualization?

Data visualization is the graphical representation of information and data. By using visual elements like charts, graphs, and maps, data visualization tools provide an accessible way to see and understand trends, outliers, and patterns in data.

Key Goals:

- **Clarity:** Make complex data understandable.
- **Insight:** Reveal hidden patterns and relationships.
- **Communication:** Share findings with others effectively.



Why is Data Viz Important?

The Power of Visuals

Our brains are wired to process visual information. We can process images almost instantly, far faster than reading text.

Good visualization leverages **pre-attentive attributes**—like color, size, and shape—to convey information at a glance, before we even actively "pay attention."

Comparing Preattentive Attributes



Megan Spengler

Sr. Visual Analytics Associate



SHAPE



COLOR



SIZE



ENCLOSURE



POSITION

Playfair +

What is Tableau?

What is Tableau?

Tableau is a leading data visualization tool that allows you to connect to, visualize, and share your data. It's known for its user-friendly drag-and-drop interface.

Focus: Tableau Public

This course focuses on **Tableau Public**, the free version. It's a powerful tool for learning, creating, and sharing your visualizations with the world. All exercises in this course are designed for Tableau Public.



What You Will Learn (Beginner Module)

- ✓ **Connect** to data sources (CSVs, Excel).
- ✓ **Understand** the Tableau Public workspace.
- ✓ **Build** fundamental charts (bars, lines, pies).
- ✓ **Filter, sort, and group** your data.
- ✓ **Create** basic calculated fields.
- ✓ **Build and share** an interactive dashboard.

The Tableau Public Interface

The Tableau Workspace

The Tableau interface is divided into key areas. Understanding these will speed up your workflow significantly. When you open Tableau and connect to data, this is the main screen you will use.



Workspace Components Explained

1. Data Pane (Your "Pantry")

- Contains all your data fields, organized into Dimensions & Measures.

2. Shelves (Your "Recipe")

- Drag fields here to build the viz.
- **Columns:** Creates columns (X-axis).
- **Rows:** Creates rows (Y-axis).

3. Canvas (Your "Plate")

- Where the visualization is built.

4. Marks Card

- Controls the visual details (color, size, label, tooltip, chart type).

5. "Show Me" Panel

- A guide that suggests chart types based on the data you've selected.

6. Toolbar

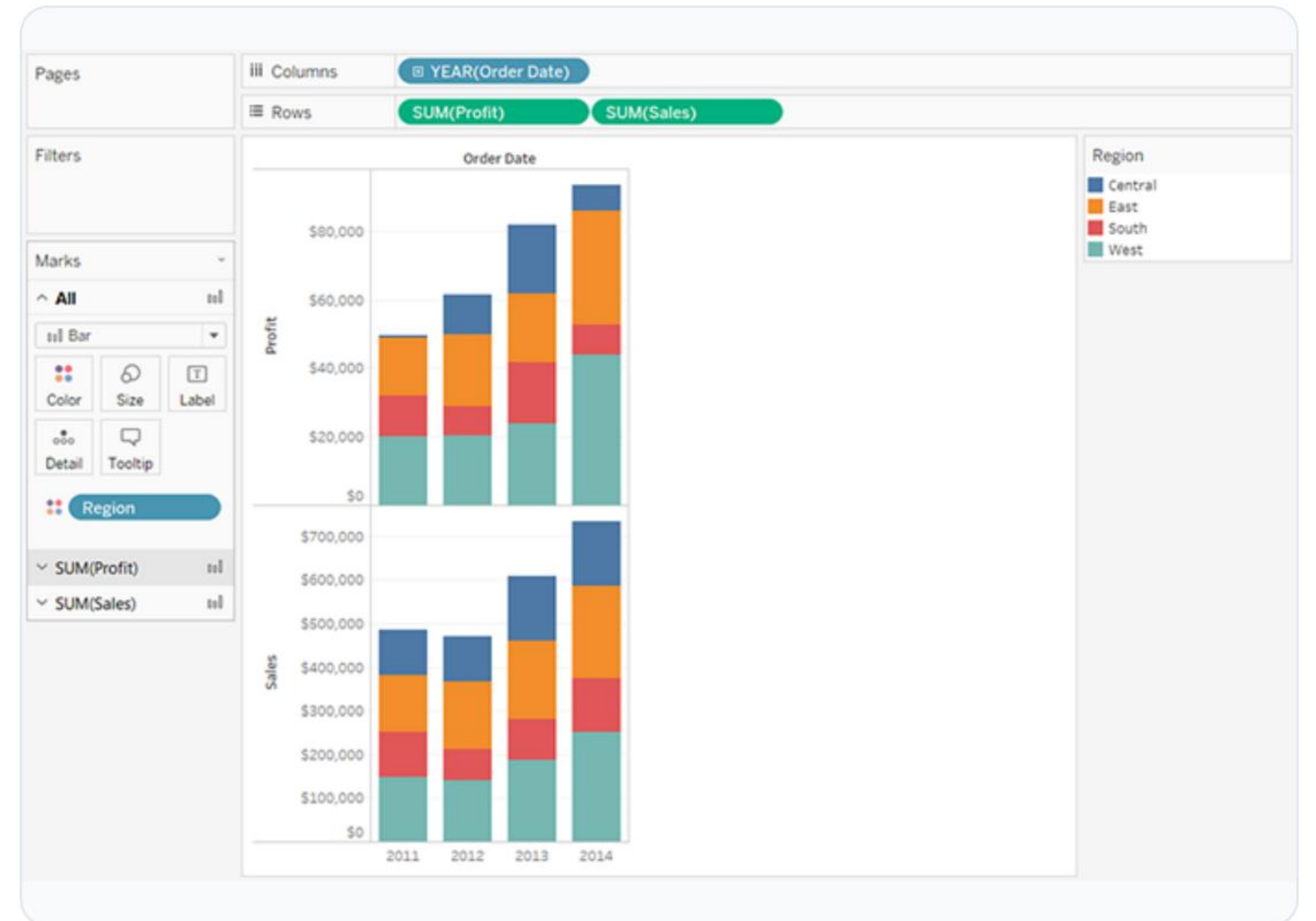
- Contains save, undo, redo, and other controls.

The "Show Me" Panel

Your Visualization Assistant

"Show Me" is your assistant. Select the data fields you're interested in from the Data Pane (e.g., hold Ctrl and click one date, one number).

It will then highlight recommended chart types. It's a great way to learn what's possible and to build charts quickly.



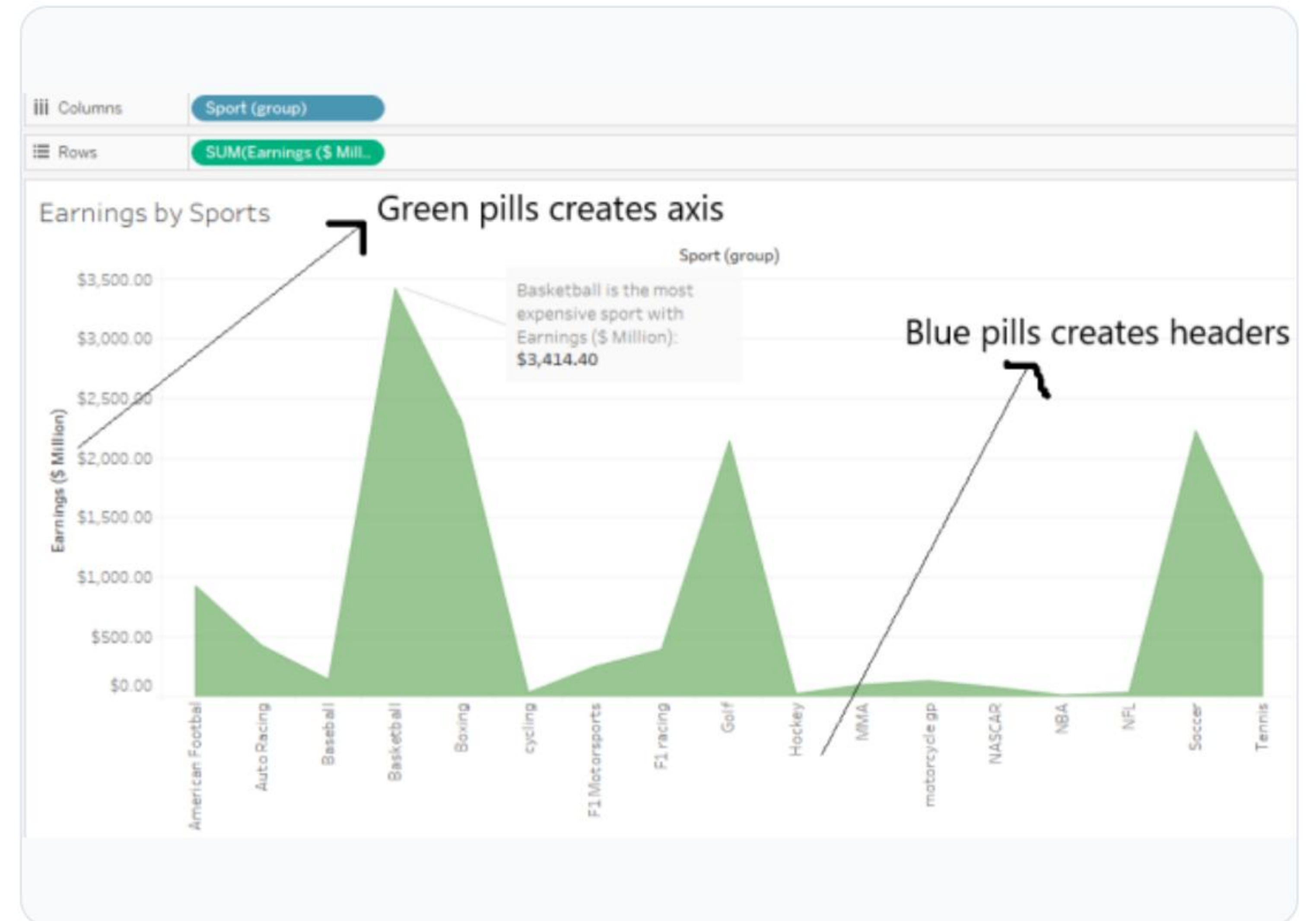
Dimensions vs. Measures (Critical Concept)

Dimensions (Blue Pills)

- **What:** Qualitative, categorical data.
- **Examples:** Department , Name , Region , Date .
- **Function:** They "slice" the data. They set the level of detail.

Measures (Green Pills)

- **What:** Quantitative, numerical data.
- **Examples:** Salary , Sales_Amount , Quantity .
- **Function:** They "fill" the data. They are aggregated (e.g., SUM, AVG).



Module 2: Connecting to Data

Using the `hr_analytics.csv` and `sales_data.csv` files.

Common Data Types in Tableau

 **Text (String):** "Sales", "Casey Gillespie"

 **Date:** 1/15/2023

 **Date & Time:** 1/15/2023 10:30:00 AM

 **Numeric (Decimal):** 123.45

 **Numeric (Whole):** 123

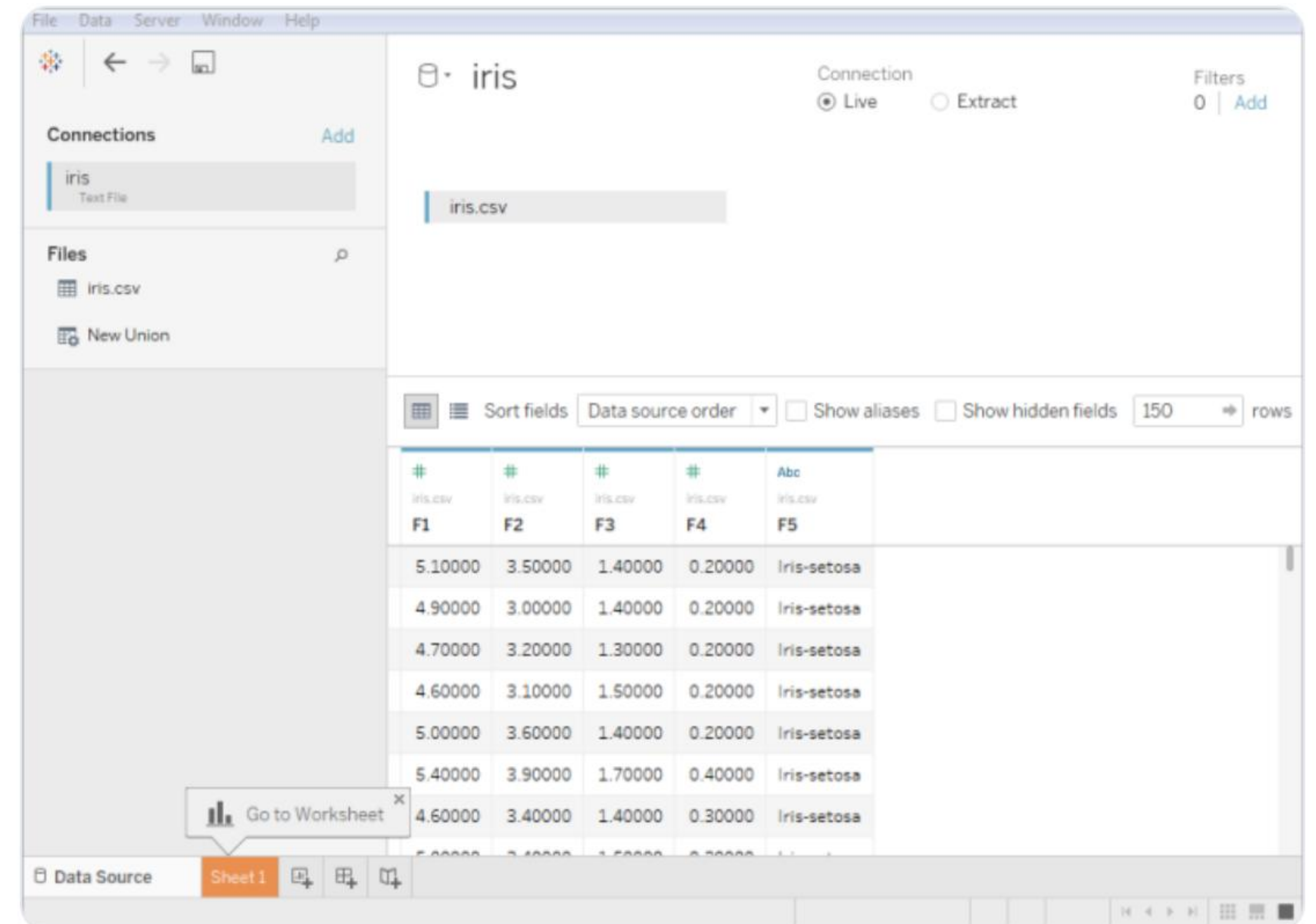
 **Boolean (T/F):** True, False, "Yes", "No"

 **Geographic:** "City", "Country" (generates maps)

Connecting to Our CSVs

Connecting to a Text File

1. Open Tableau Public.
2. In the "Connect" pane on the left, click on **"Text File"**.
3. Navigate to and select your `hr_analytics.csv` file.
4. Tableau will open the Data Source page.
5. To add the second file, click "Add" next to Connections, select "Text File" again, and choose `sales_data.csv` .

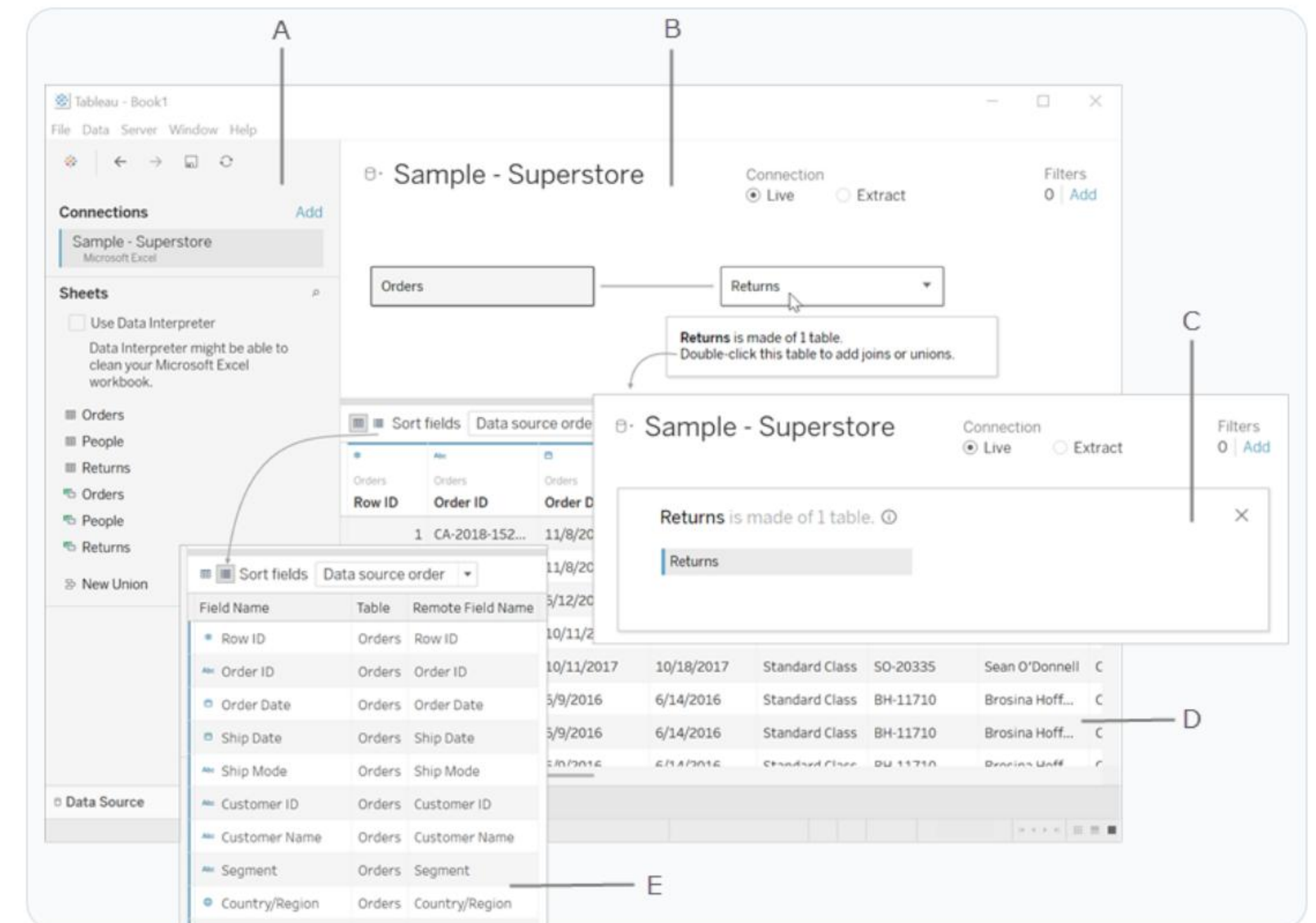


The Data Source Page

Preview Your Data

This page shows a preview of your data. Here you can:

- Verify your data has loaded correctly.
- Check and change data types (e.g., changing a number stored as text).
- Join or Union data (an intermediate topic).
- See metadata (field names).



Let's Start Building!

”

Once your data is loaded, click the "Sheet 1" tab at the bottom of the window. This will take you to the main worksheet.

”

You will now see your data fields (Dimensions and Measures) in the Data Pane.



Module 3: Building Basic Charts

From data to insight in just a few clicks.

The Bar Chart

What is it?

Bar charts are the most common chart type. They are used to compare a measure across different categories (dimensions).

Best for: Comparing values (e.g., Sales by Region, Employees by Department).

How to Build:

1. Drag a **Dimension** (e.g., `Department`) to **Columns**.
2. Drag a **Measure** (e.g., `Employee_ID`) to **Rows**.
3. Tableau will default to `SUM(Employee_ID)` . This is wrong!
4. Change this by right-clicking the green pill and selecting **Measure > Count (Distinct)**.

Exercise 1: Employee Count

EX. 1

Bar Chart

Exercise: Employee Count by Department

Dataset: `hr_analytics.csv`

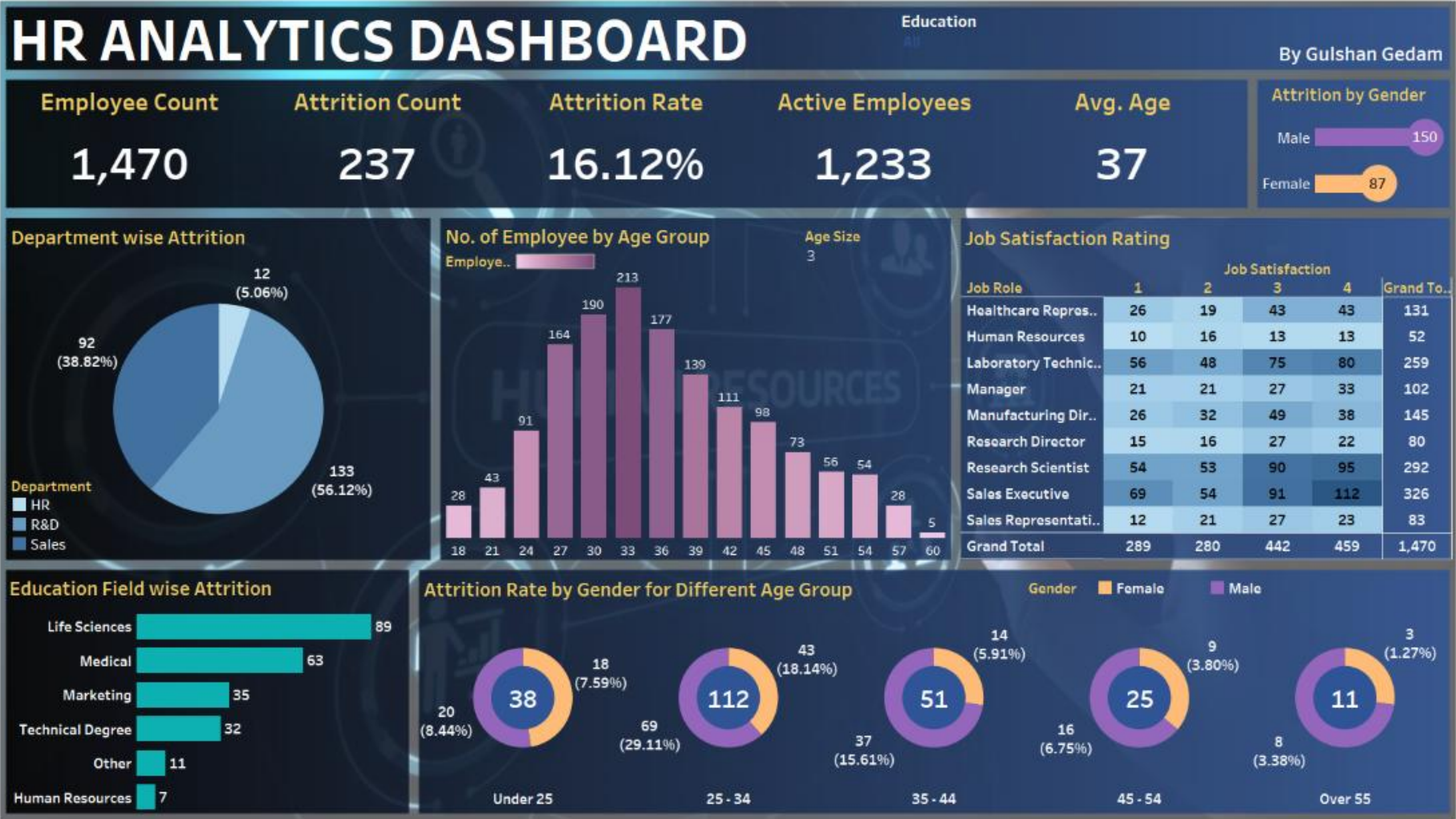
Task: Create a horizontal bar chart showing the total number of employees in each department. (Hint: You will need to use a distinct count of `Employee_ID`).

Solution: Exercise 1

Employee Count by Department

Your chart should look like this. The "IT" and "Sales" departments have the most employees.

Tip: Click the "Swap Rows and Columns" button in the toolbar to make it horizontal.



The Line Chart

What is it?

Line charts are used to track a measure's performance over time (a date dimension).

Best for: Seeing trends, seasonality, or changes over time.

How to Build:

1. Drag a **Date Dimension** (e.g., `Date`) to **Columns**.
2. Drag a **Measure** (e.g., `Sales_Amount`) to **Rows**.
3. Tableau will automatically create a line chart. You can use the "+" on the `YEAR(Date)` pill to drill down.

Exercise 2: Sales Over Time

Ex. 2

Line Chart

Exercise: Sales Over Time

Dataset: `sales_data.csv`

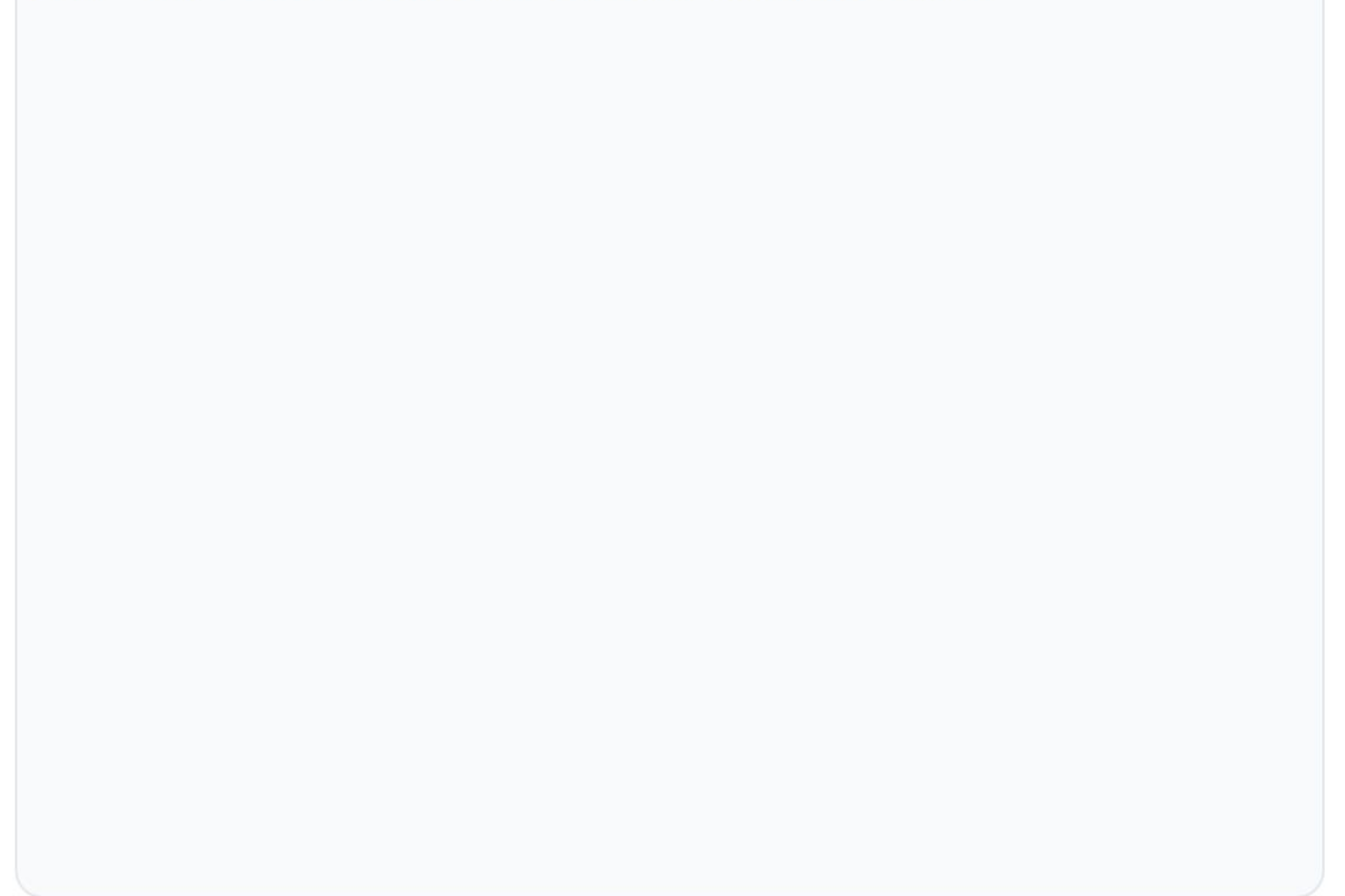
Task: Create a line chart showing the `SUM(Sales_Amount)` by `Date`. Drill down to the **Quarter** level.

Solution: Exercise 2

Sales by Quarter

This chart shows the sales trend. You can see peaks and valleys in different quarters. It looks like sales consistently peak in Q4.

 Line chart created in Tableau showing sum of sales by quarter



The Pie Chart

What is it?

Pie charts show part-to-whole relationships. They represent a measure's total, broken down by a dimension.

Best for: Showing proportions (e.g., % of total).

Warning: Use them sparingly! They are hard to read with more than 2-3 categories. A bar chart is often better.

How to Build:

1. Select a **Dimension** and a **Measure** (e.g., `Gender` and `Employee_ID`).
2. Click the "Show Me" panel and select the Pie Chart.
3. Drag `Gender` to **Color**.
4. Drag `COUNTD(Employee_ID)` to **Angle**.

Exercise 3: Employee Gender

EX. 3

Pie Chart

Exercise: Employee Gender Ratio

Dataset: `hr_analytics.csv`

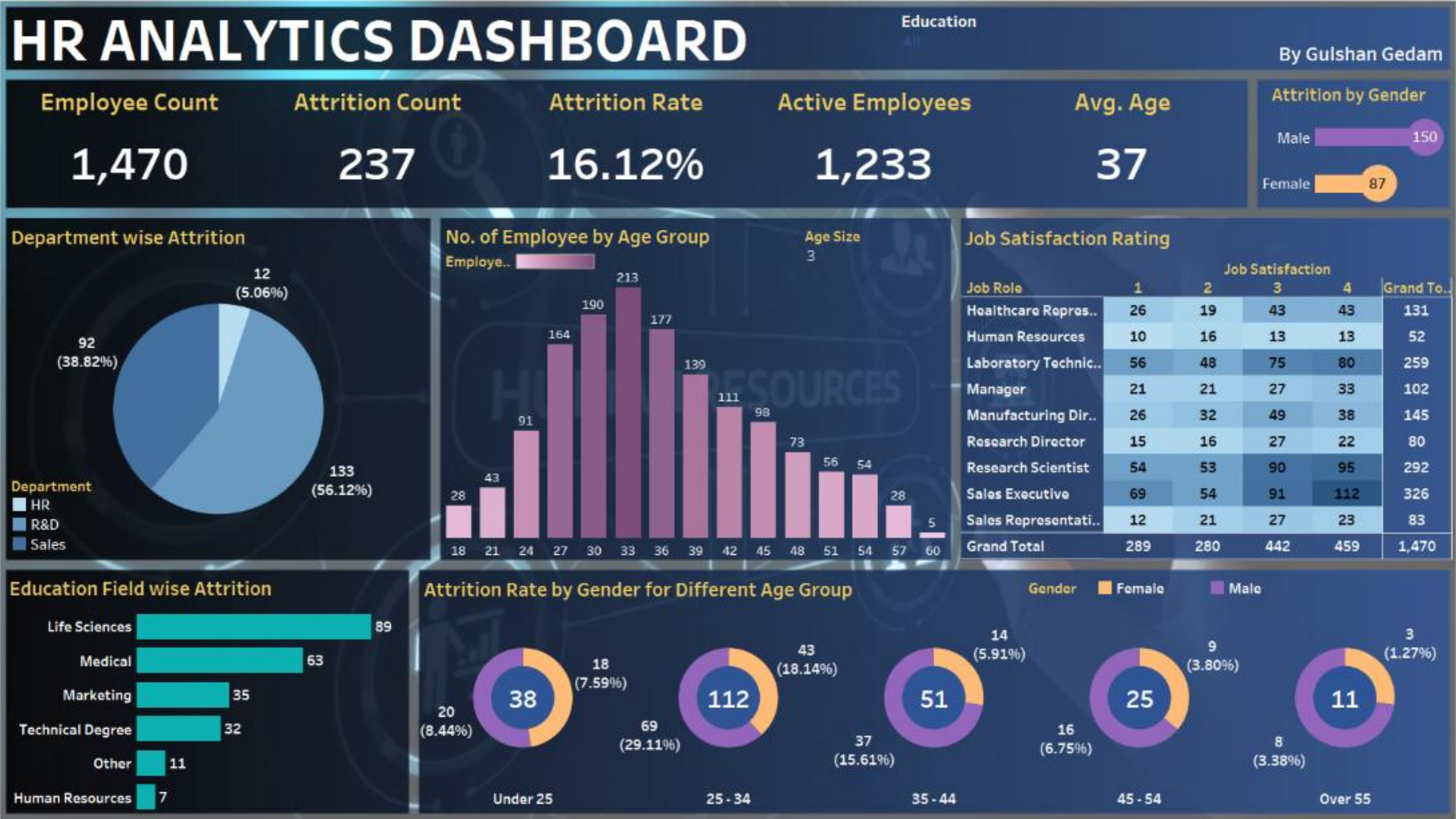
Task: Create a pie chart showing the proportion of employees by Gender . Add labels to show the percentage.

Solution: Exercise 3

Employee Gender Ratio

Your pie chart should clearly show the mix of male and female employees.

Tip: To add percent labels, drag `COUNTD(Employee_ID)` to Label, then right-click the pill > Quick Table Calculation > Percent of Total.



Module 4: Enhancing Your Visuals

Adding context and detail with filters, colors, and sorting.

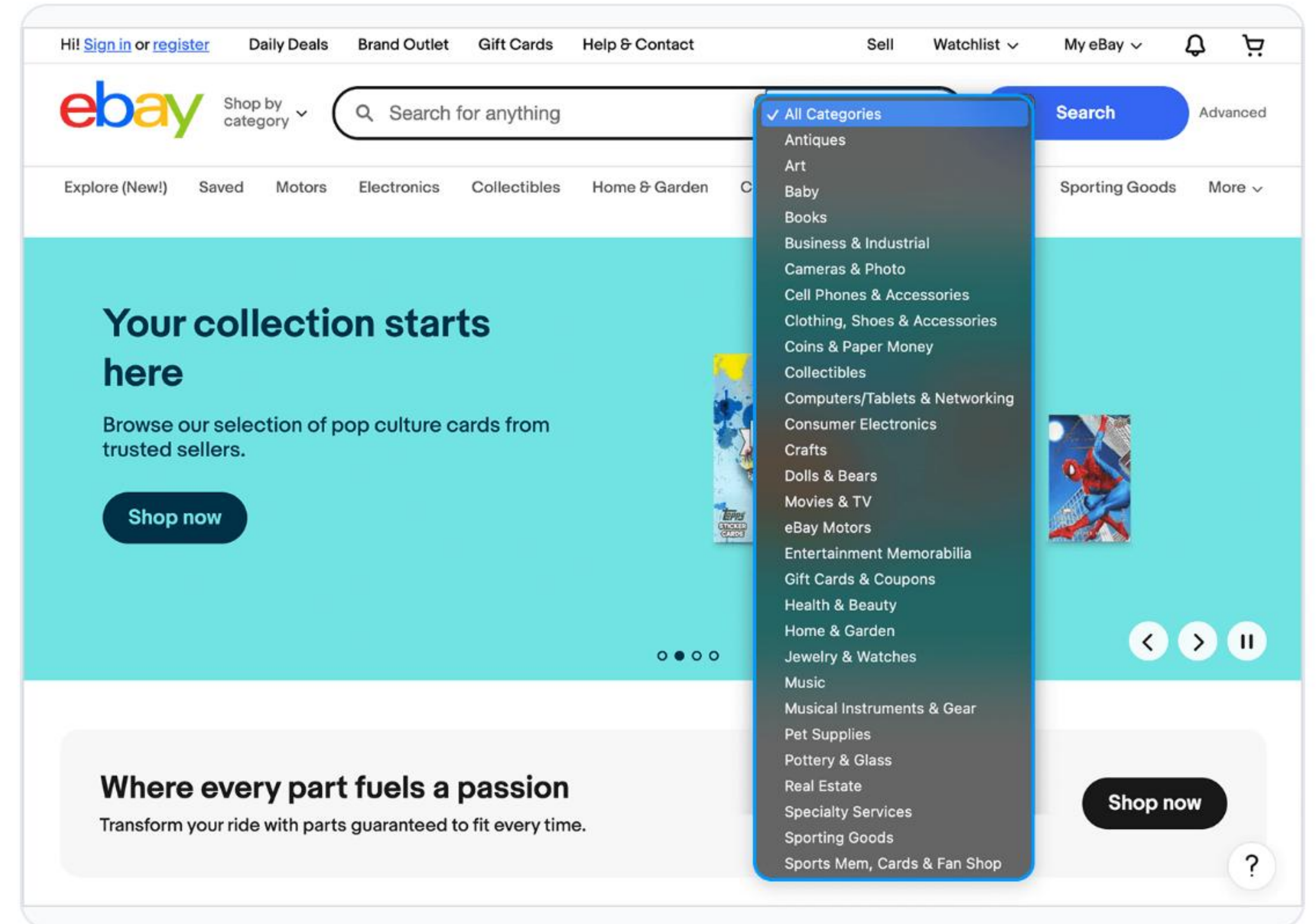
What are Filters?

What are Filters?

Filters allow you to include or exclude data from your view. They are essential for letting users explore the data or for focusing on a specific part.

How to Use:

1. Drag any field (e.g., `Region`) to the **"Filters"** shelf.
2. A dialog box will open.
3. Check the boxes you want to include.
4. To make it interactive, right-click the `Region` pill on the Filters shelf and select **"Show Filter"**.



Exercise 4: Filter by Category

Ex. 4

Filters

Exercise: Filter by Region

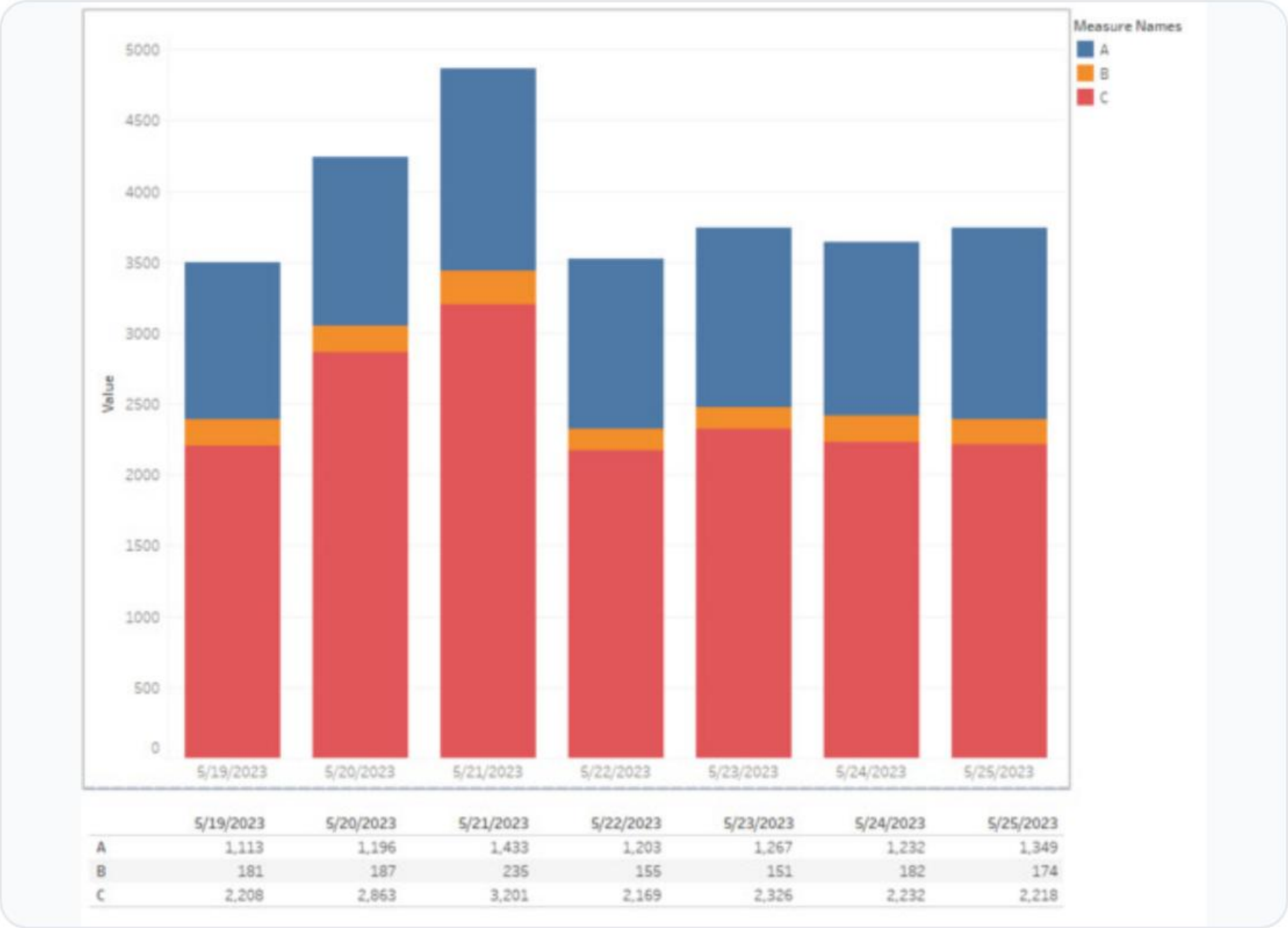
Dataset: `sales_data.csv`

Task: Create a bar chart of `SUM(Sales_Amount)` by `Product_Category`. Then, add a filter for `Region` and show **only** the "North America" region.

Solution: Exercise 4

North America Sales

Your view is now filtered. The chart only shows sales from North America.



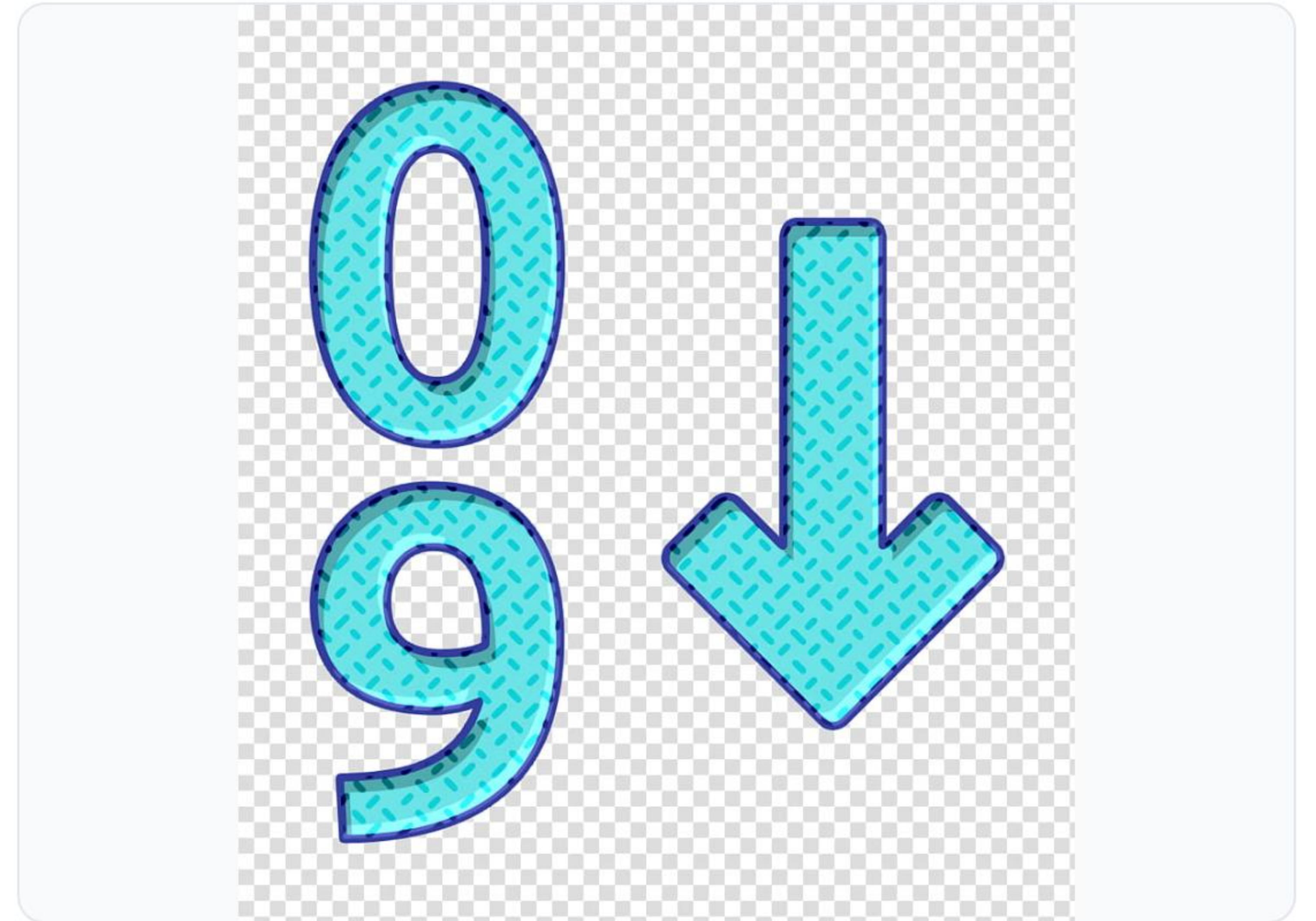
Sorting Data

Sorting Data

By default, Tableau sorts alphabetically. You usually want to sort by a measure to see the highest or lowest values.

How to Sort:

- **Simple Sort:** Click the sort icons on the axis or toolbar.
- **Manual Sort:** Right-click the blue dimension pill (e.g., Department) and select "Sort".
- In the dialog, choose "**Sort by Field**". Set Sort Order to "Descending" and Field Name to your measure.



Exercise 5: Sorting

EX. 5

Sorting

Exercise: Top Paid Departments

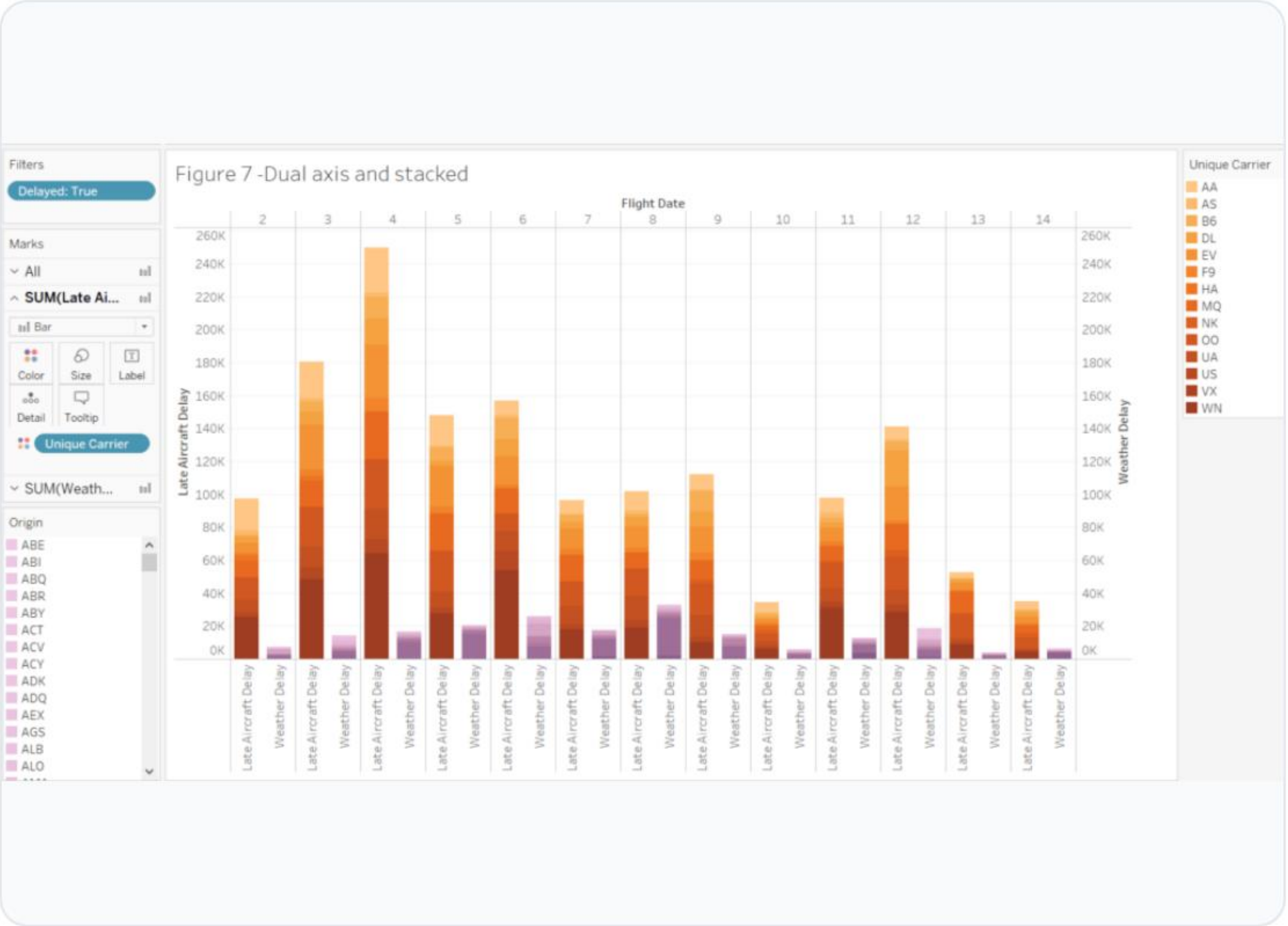
Dataset: `hr_analytics.csv`

Task: Create a bar chart of Department by `AVG(Salary)`. Sort it from **highest** average salary to **lowest**.

Solution: Exercise 5

Top Paid Departments

Now you can easily see which departments have the highest average salary. "Finance" and "Marketing" lead the way.



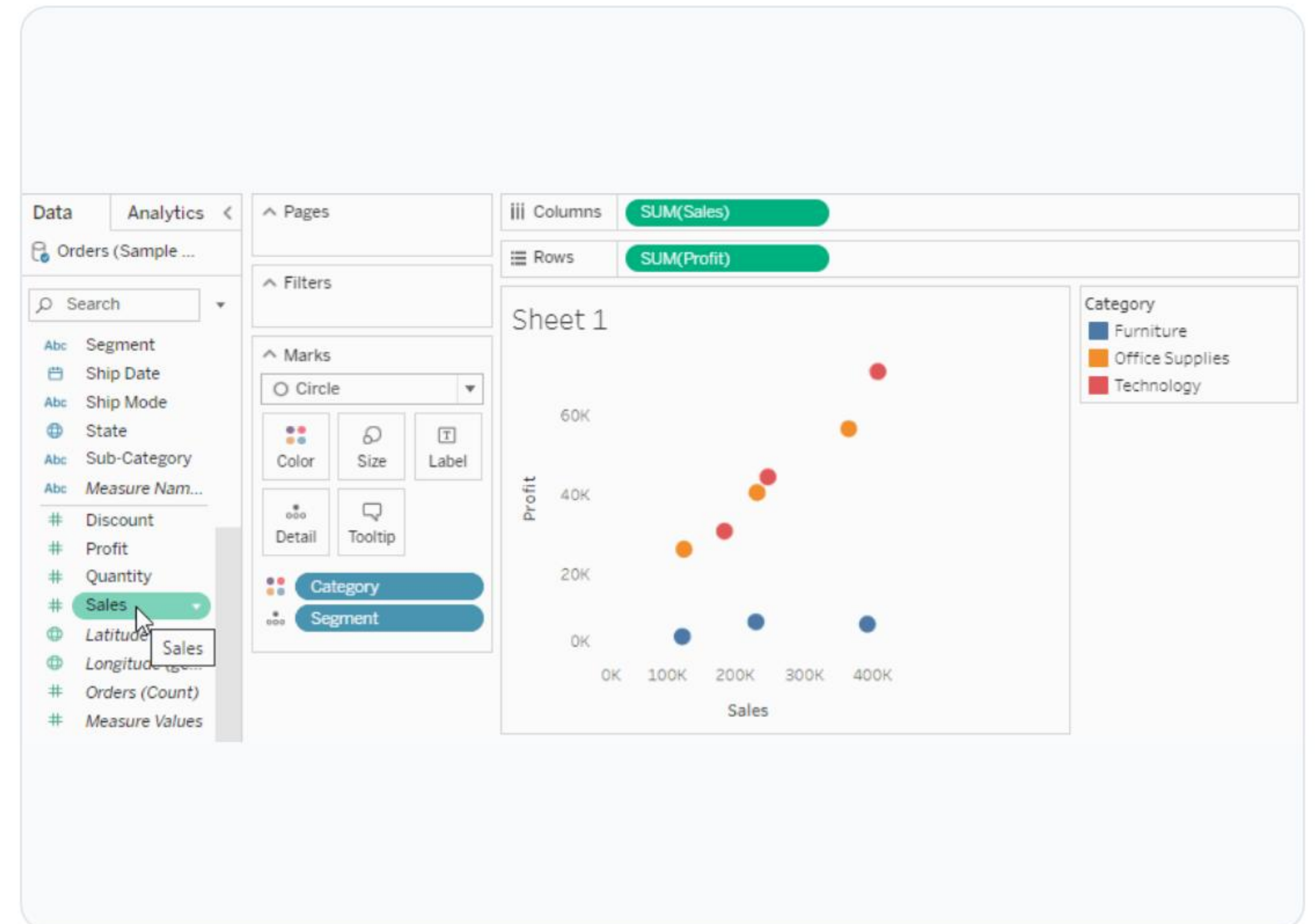
The Marks Card: Color

Using Color

Color is one of the most powerful visual cues. You can use it to break down your chart by another dimension.

How to Use:

1. Create a bar chart of `SUM(Sales_Amount)` by `Region`.
2. Drag the `Product_Category` dimension onto the **"Color"** button in the Marks Card.
3. Your single bar for each region will now become a stacked bar.



Exercise 6: Colors

Ex. 6

Colors

Exercise: Sales Mix by Region

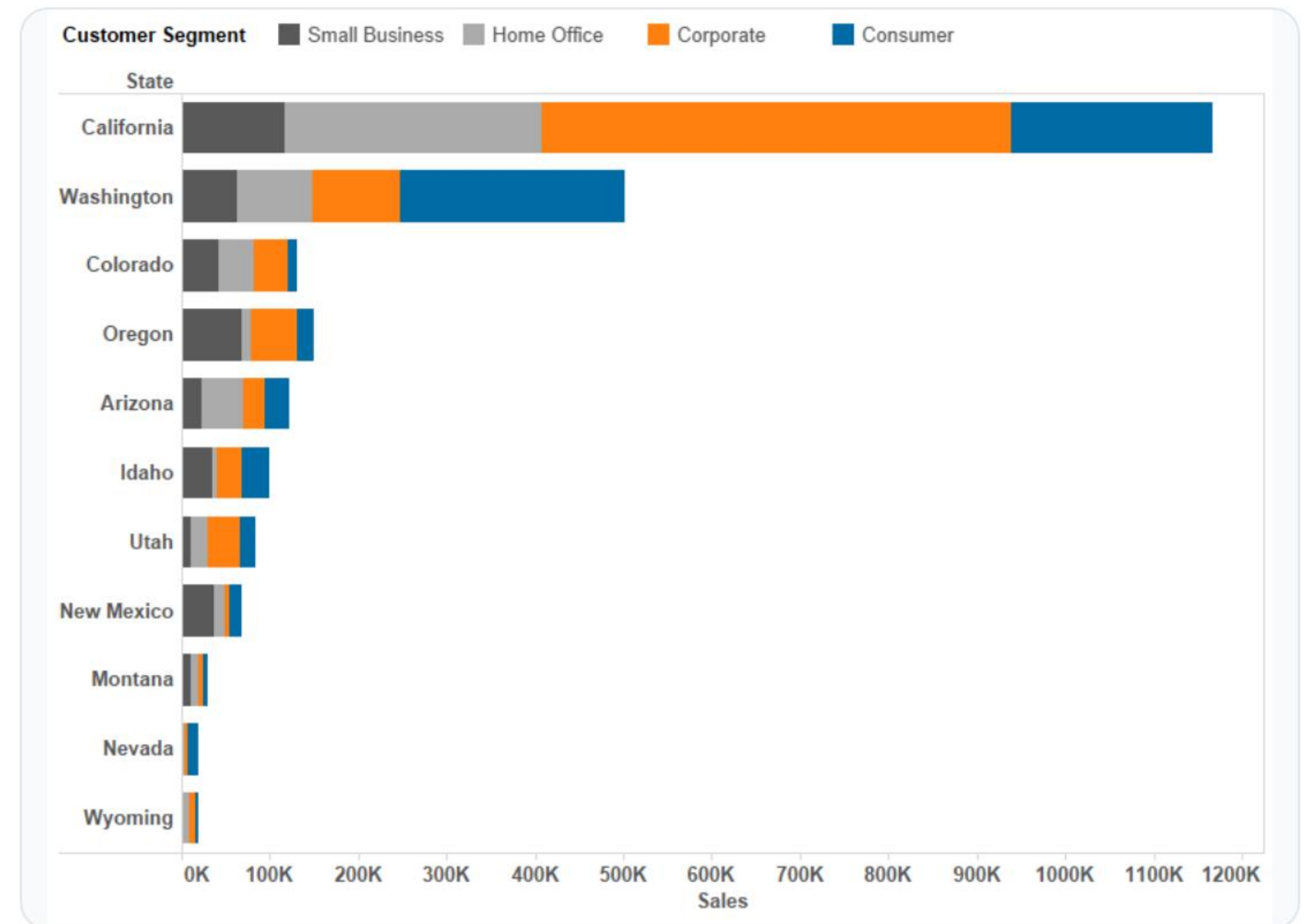
Dataset: `sales_data.csv`

Task: Create a stacked bar chart showing `SUM(Sales_Amount)` by `Region`, with the bars broken down by `Product_Category` (using `color`).

Solution: Exercise 6

Sales Mix by Region

This chart is much richer. You can now see *what* is selling in each region, not just the total. For example, "Electronics" is a huge part of sales in Asia Pacific.



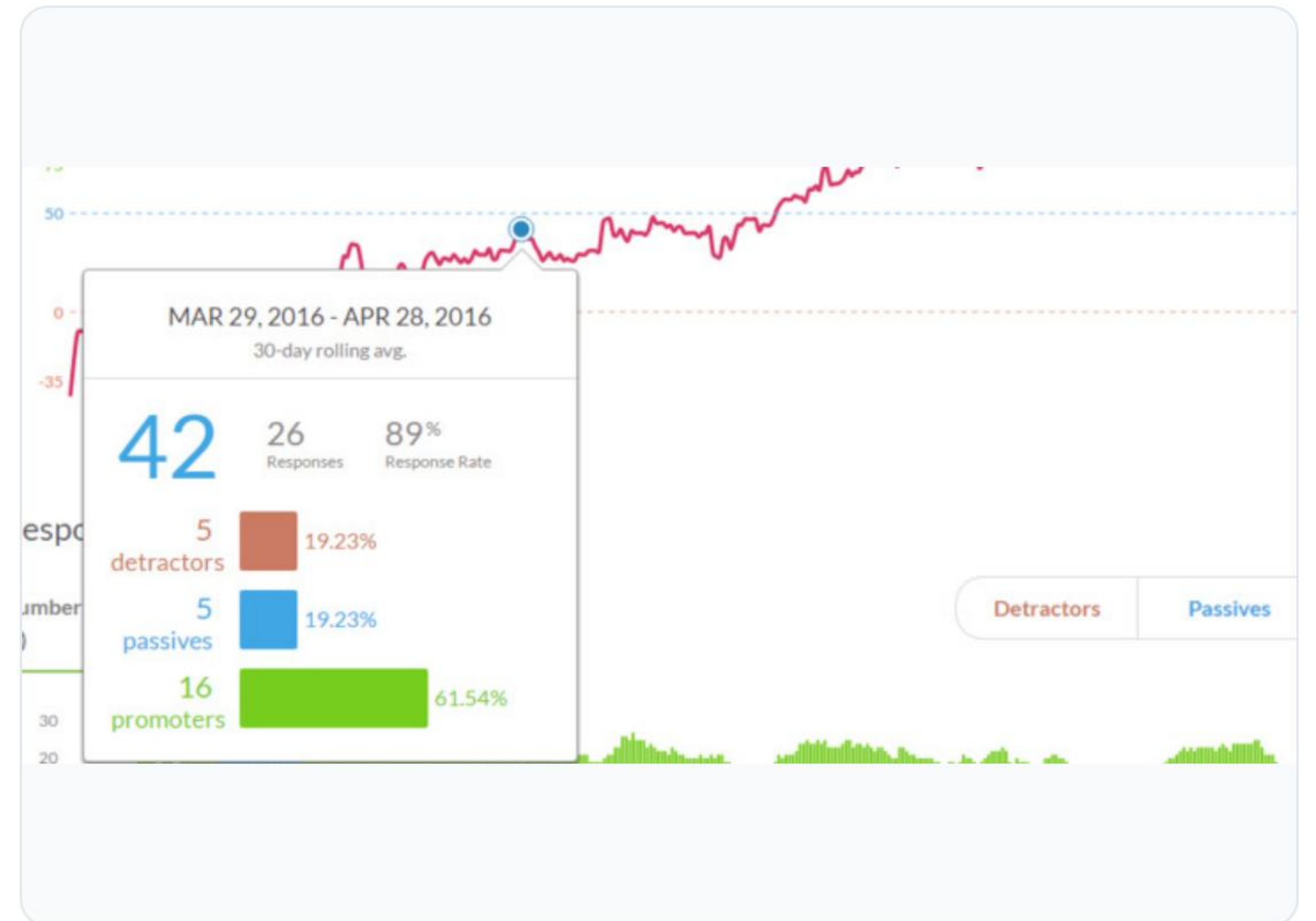
Editing Tooltips

What are Tooltips?

Tooltips are the "pop-up" boxes that appear when you hover over a data point. By default, they show the data used in the chart.

You can edit them by clicking the **"Tooltip"** button in the Marks Card.

You can add more fields (just drag them onto the Tooltip button) or type static text to make them more readable (e.g., "Region: ").



Adding Labels

What are Labels?

Sometimes you want the values to be directly visible on the chart without hovering.

Warning: Don't clutter your chart! This is best used for charts with few data points, like a bar chart with 5-6 bars.

How to Add Labels:

1. Drag the measure you want to display (e.g., `SUM(Sales_Amount)`) onto the "**Label**" button in the Marks Card.
2. The values will now appear on your chart.
3. You can format these labels by clicking the "Label" button.

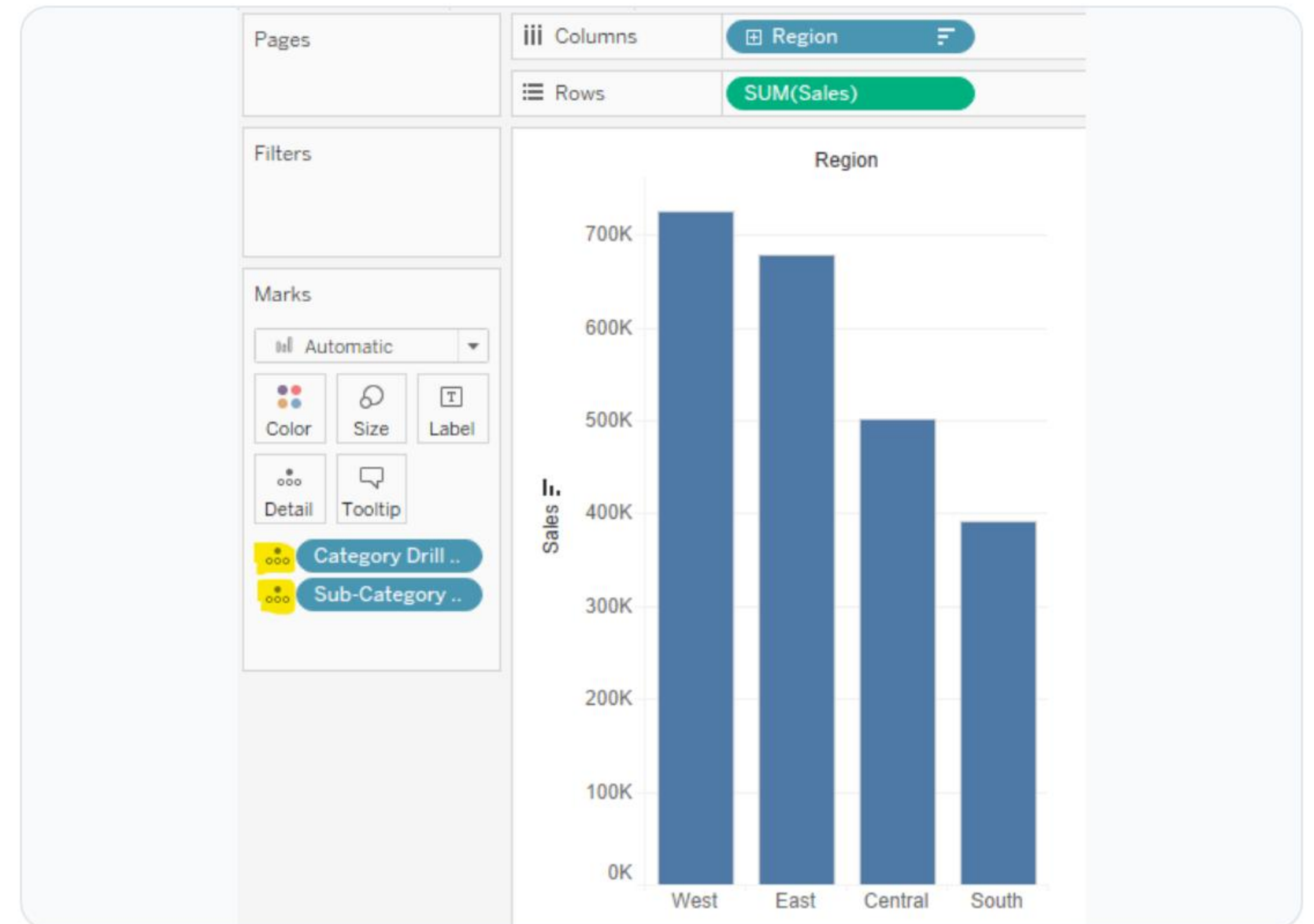
Understanding Hierarchies

What are Hierarchies?

Hierarchies allow you to "drill down" into your data. Tableau creates them automatically for dates (Year > Quarter > Month > Day).

You can create your own by dragging one dimension onto another in the Data Pane.

Example: Drag `Product_Name` onto `Product_Category` to create a `Category > Product` hierarchy.



Exercise 7: Hierarchies

Ex. 7

Hierarchies

Exercise: Date Drill-Down

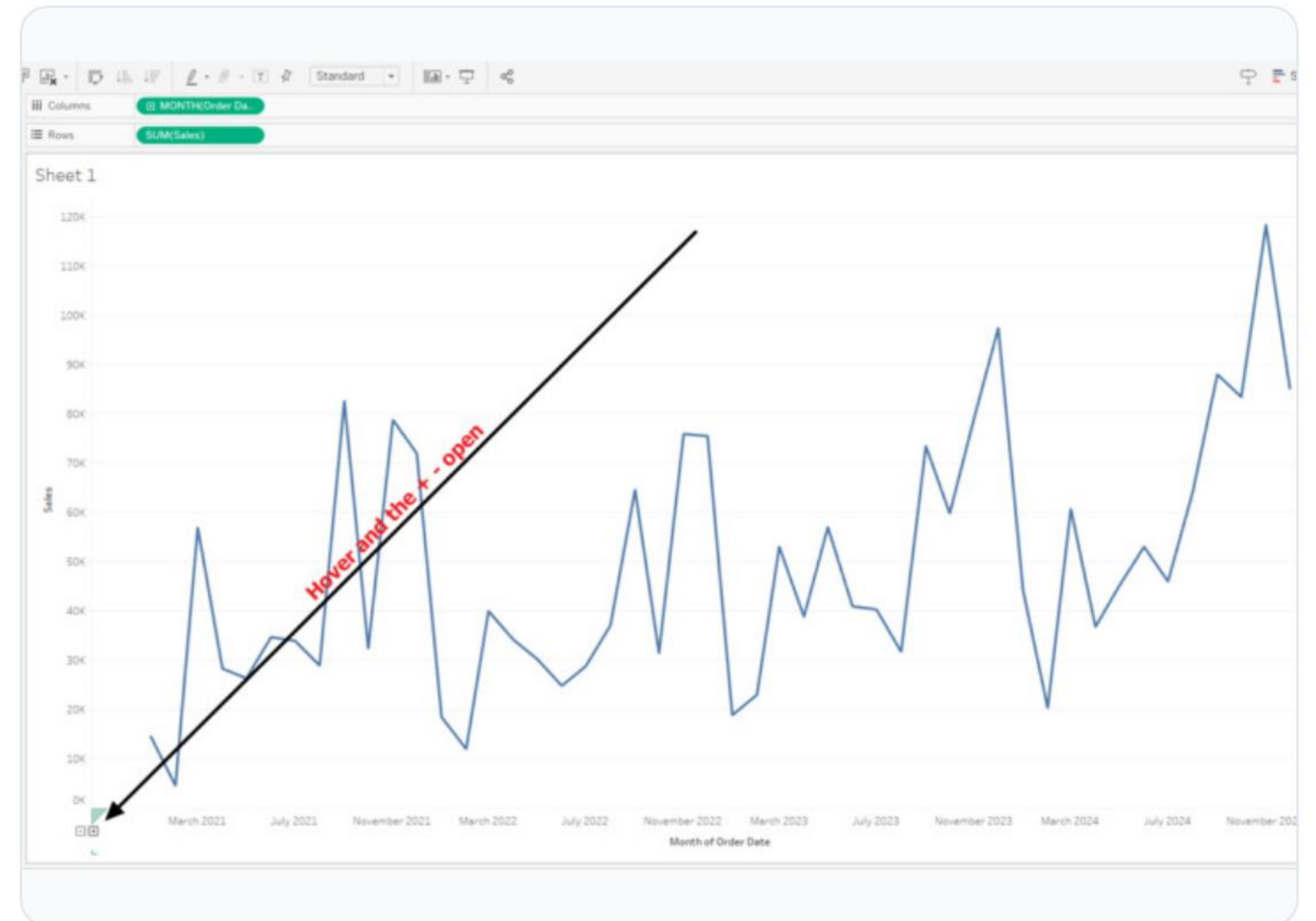
Dataset: `sales_data.csv`

Task: Use your "Sales Over Time" line chart from Exercise 2. It should be at the `YEAR` level. Click the "+" sign on the `YEAR(Date)` pill in the Columns shelf. What happens? Click it again.

Solution: Exercise 7

Drilling Down

Clicking the "+" on the date pill drills you down to the next level (Year > Quarter > Month). This is a fundamental way to explore time-based data.



Module 5: Calculated Fields

Creating new data from your existing data.

What are Calculated Fields?

Calculated Fields (fx)

A calculated field is a **new field** (either a dimension or measure) that you create using a formula. This is how you create new metrics that don't exist in your original dataset.

How to Create:

- Click the down-arrow at the top of the Data Pane and select "Create Calculated Field..."
- A new window will open for you to name your field and write your formula.



Example: Numeric Calculation

Example: Profit Margin

Our `sales_data.csv` has `Sales_Amount` and `Profit`. It does *not* have `Profit Margin`.

We can create it!

How to Build:

1. Create a new Calculated Field named "Profit Margin".
2. Enter the formula:
`SUM([Profit]) / SUM([Sales_Amount])`
3. This new "Profit Margin" pill will appear in the Data Pane. You can now use it like any other measure!
4. **Tip:** Right-click your new pill, go to "Default Properties" > "Number Format..." > "Percentage".

Exercise 8: Profit Margin

Ex. 8

Calculations

Exercise: Profit Margin

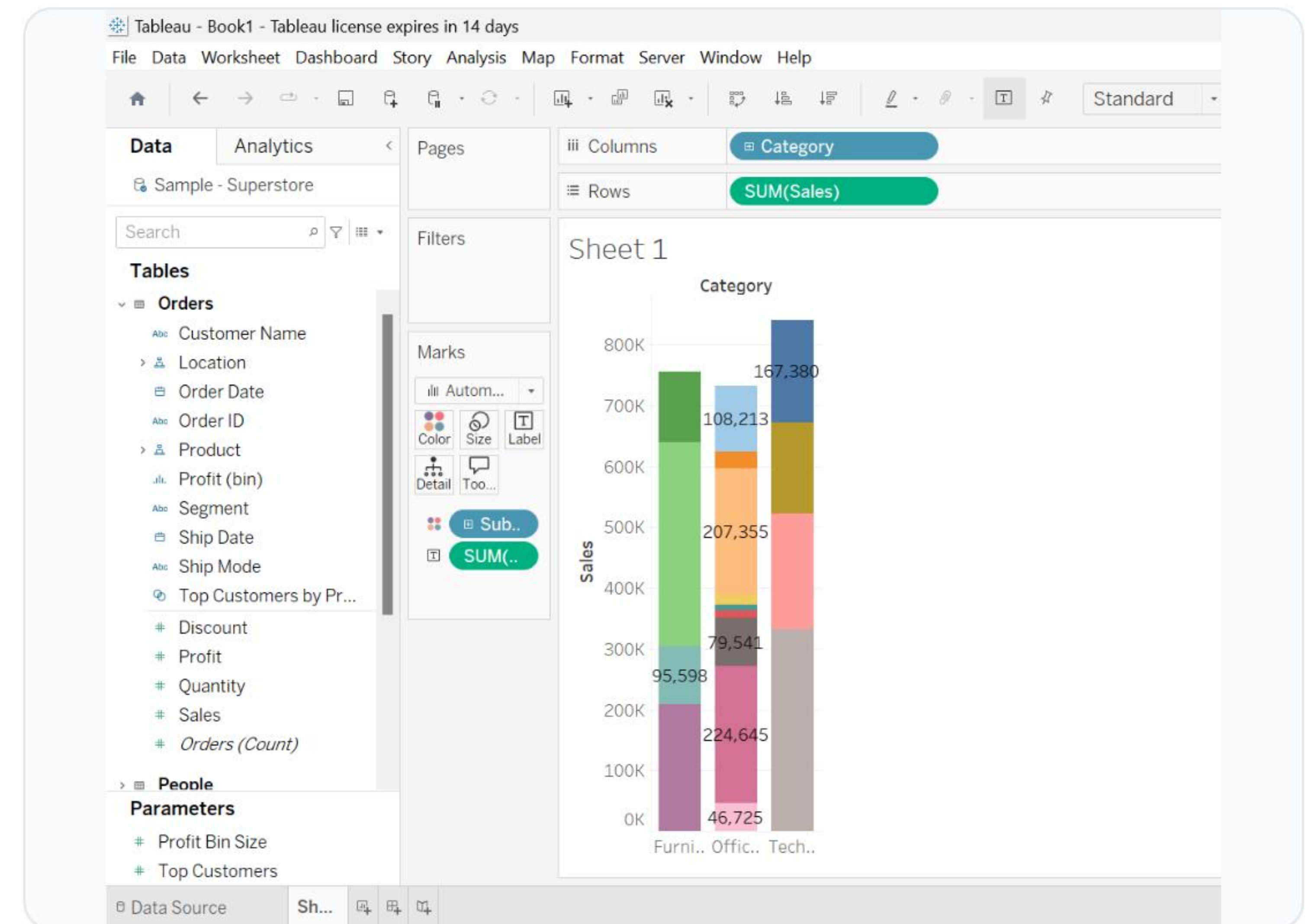
Dataset: `sales_data.csv`

Task: Create the "Profit Margin" calculated field. Then, create a bar chart showing `Product_Category` by `Profit Margin`.

Solution: Exercise 8

Profit Margin by Category

Your new calculated field allows you to see that "Electronics" and "Books" are highly profitable, while "Clothing" is less so.



Example: String Functions

Manipulating Text

You can also manipulate text. The `hr_analytics.csv` has a `Name` field, but what if you want just the last name?

Key Functions:

- `SPLIT( [string], [delimiter], [part] )`
`SPLIT( "Casey Gillespie", " ", 2 ) = "Gillespie"`
- `LEFT( [string], [num_chars] )`
- `RIGHT( [string], [num_chars] )`
- `UPPER( [string] ) / LOWER( [string] )`

Exercise 9: String Function

Ex. 9

Calculations

Exercise: Get Last Name

Dataset: `hr_analytics.csv`

Task: Create a calculated field named "Last Name" that extracts the last name from the `Name` field. (Hint: Use `SPLIT`).

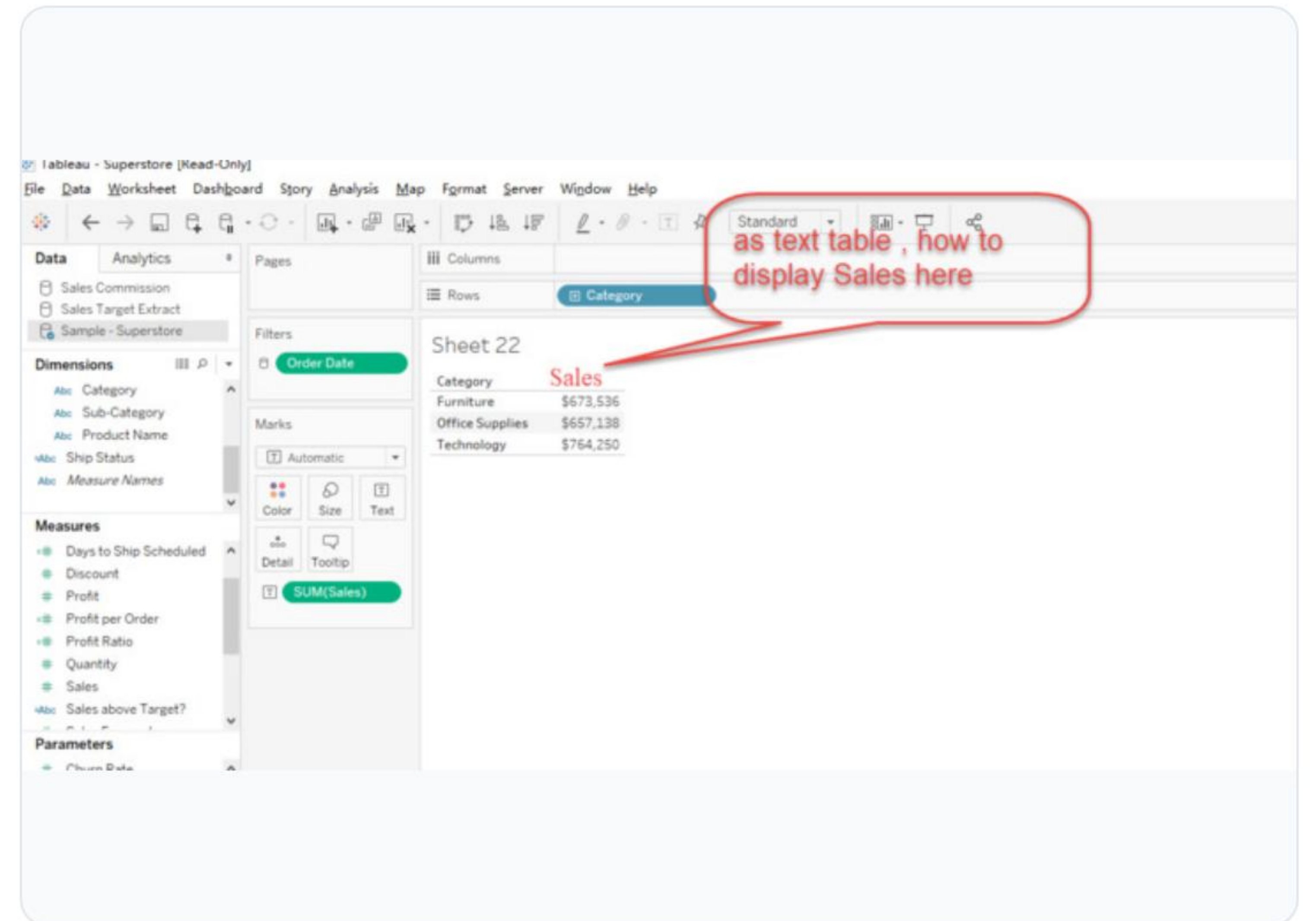
Solution: Exercise 9

Solution: Get Last Name

Formula:

```
SPLIT( [Name], " ", 2 )
```

(Note: This simple split works for names like "Casey Gillespie" but may fail on names with middle names or suffixes like "DVM". This is a real-world data cleaning challenge!)



Example: Date Functions

Working with Dates

Date functions are extremely powerful for analysis, like calculating the time between two events.

Key Functions:

- `DATEDIFF( [date_part], [start_date], [end_date] )`
e.g., `DATEDIFF('year', [Hire_Date], TODAY())` = Tenure
- `YEAR( [date] )`
- `MONTH( [date] )`
- `TODAY()` or `NOW()`

Exercise 10: Date Function

Ex. 10

Calculations

Exercise: Employee Tenure

Dataset: `hr_analytics.csv`

Task: Create a calculated field "Years at Company". (Hint: Use `DATEDIFF` between `Hire_Date` and `TODAY()` .You'll want the 'year' part.)

Module 6: Dashboards & Sharing

Bringing it all together to tell a story.

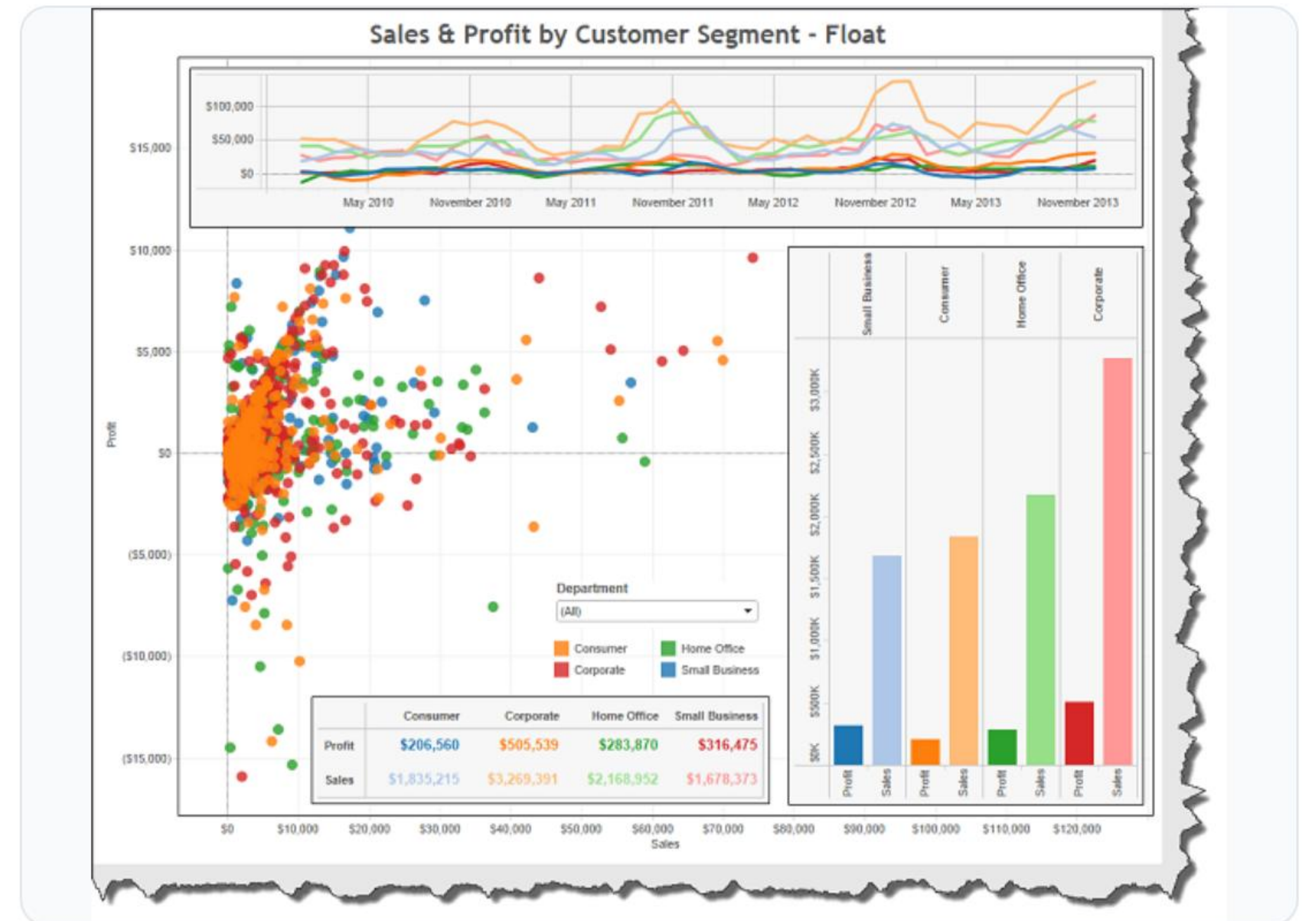
What is a Dashboard?

Worksheet vs. Dashboard

A **Worksheet** is where you build a single chart (a "viz").

A **Dashboard** is a collection of worksheets and other objects (text, images, filters) arranged on a single screen to present a complete story.

Click the "New Dashboard" icon (a grid) at the bottom, next to "Sheet 1".



Building a Dashboard

How to Build:

1. Open your new dashboard.
2. On the left, you will see a list of all your worksheets.
3. Drag your worksheets (e.g., "Dept Bar Chart", "Gender Pie Chart") onto the canvas.
4. By default, they are "Tiled," meaning they snap into place.
5. Drag your filters and legends into view.

Key Tips:

- Resize the objects by dragging the borders between them.
- Use "Floating" for objects you want to place anywhere (like a title).
- Add "Text" objects to add titles and explanations.

Making it Interactive: Actions

What are Actions?

The real power of dashboards comes from "Actions."

Goal: When you click on a department in the bar chart, you want the pie chart to filter and show **only** the gender ratio for **that** department.

How to Add a Filter Action:

1. With your dashboard open, go to **Dashboard > Actions...**
2. Click "Add Action" > "Filter..."
3. **Source Sheet:** Check your bar chart.
4. **Target Sheet:** Check your pie chart.
5. Run Action On: **Select**.
6. Click **OK**. Now try it!

Exercise 11: HR Dashboard

Ex. 11 Dashboard

Exercise: Build an HR Dashboard

Dataset: `hr_analytics.csv`

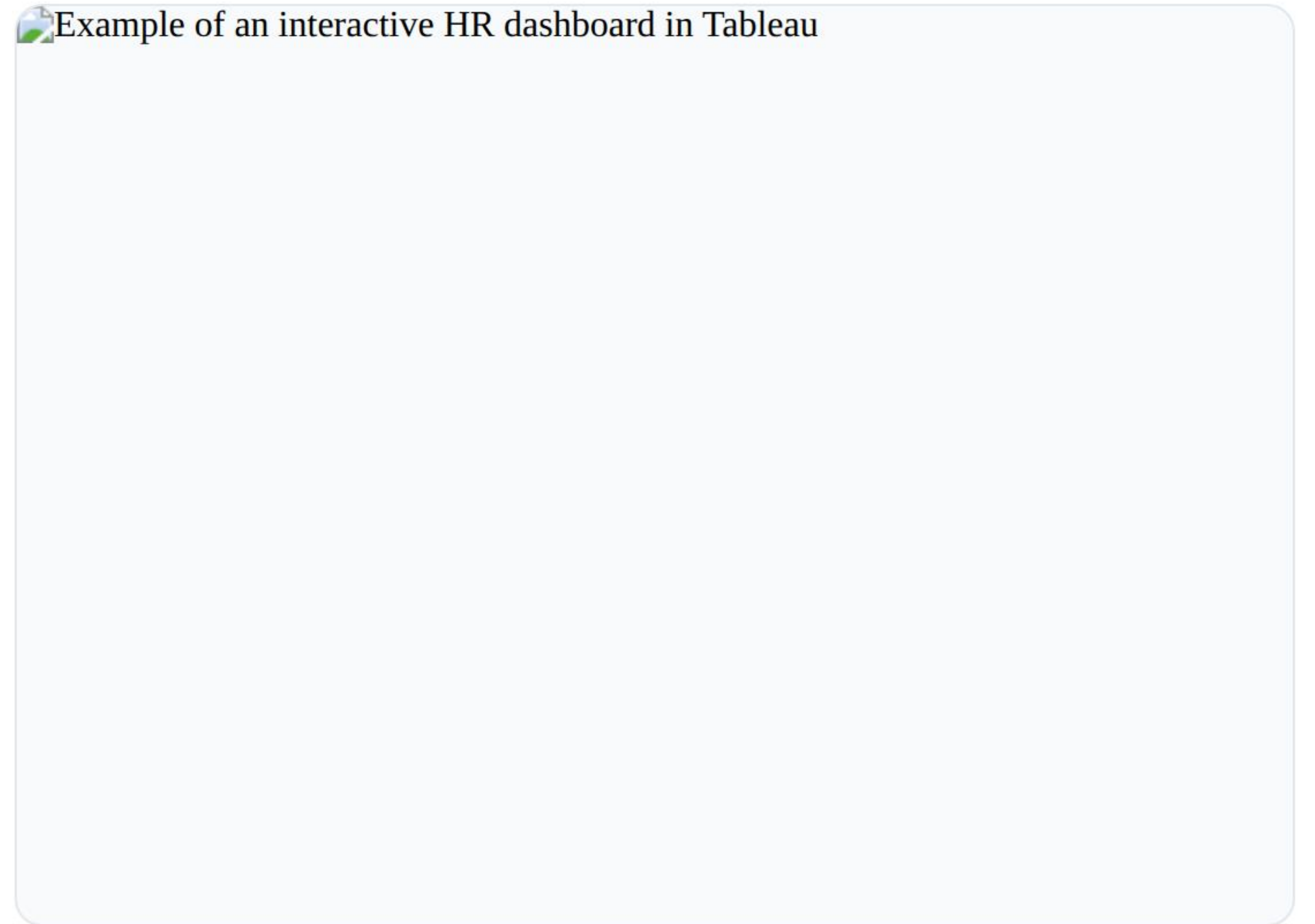
Task: Create a dashboard with your "Employee Count by Department" bar chart and your "Employee Gender Ratio" pie chart. Add a filter action so that clicking a department filters the pie chart.

Solution: Exercise 11

Interactive HR Dashboard

Your final dashboard should be interactive. This is the foundation of data analysis—allowing users to ask and answer their own questions.

 Example of an interactive HR dashboard in Tableau



Assignment 1: HR Attrition

Assignment: HR Attrition Analysis

Dataset: `hr_analytics.csv`

Goal: The CHRO wants to know why employees are leaving (Attrition = "Yes").

Tasks:

1. Find the "Attrition" field.
2. Create visuals to see if Attrition is related to: `AVG(Salary)` , `Department` , `Performance_Rating` , `Years_at_Company` .
3. Build a dashboard to present your findings.



Assignment 2: Sales Review

Assignment: Sales Performance Review

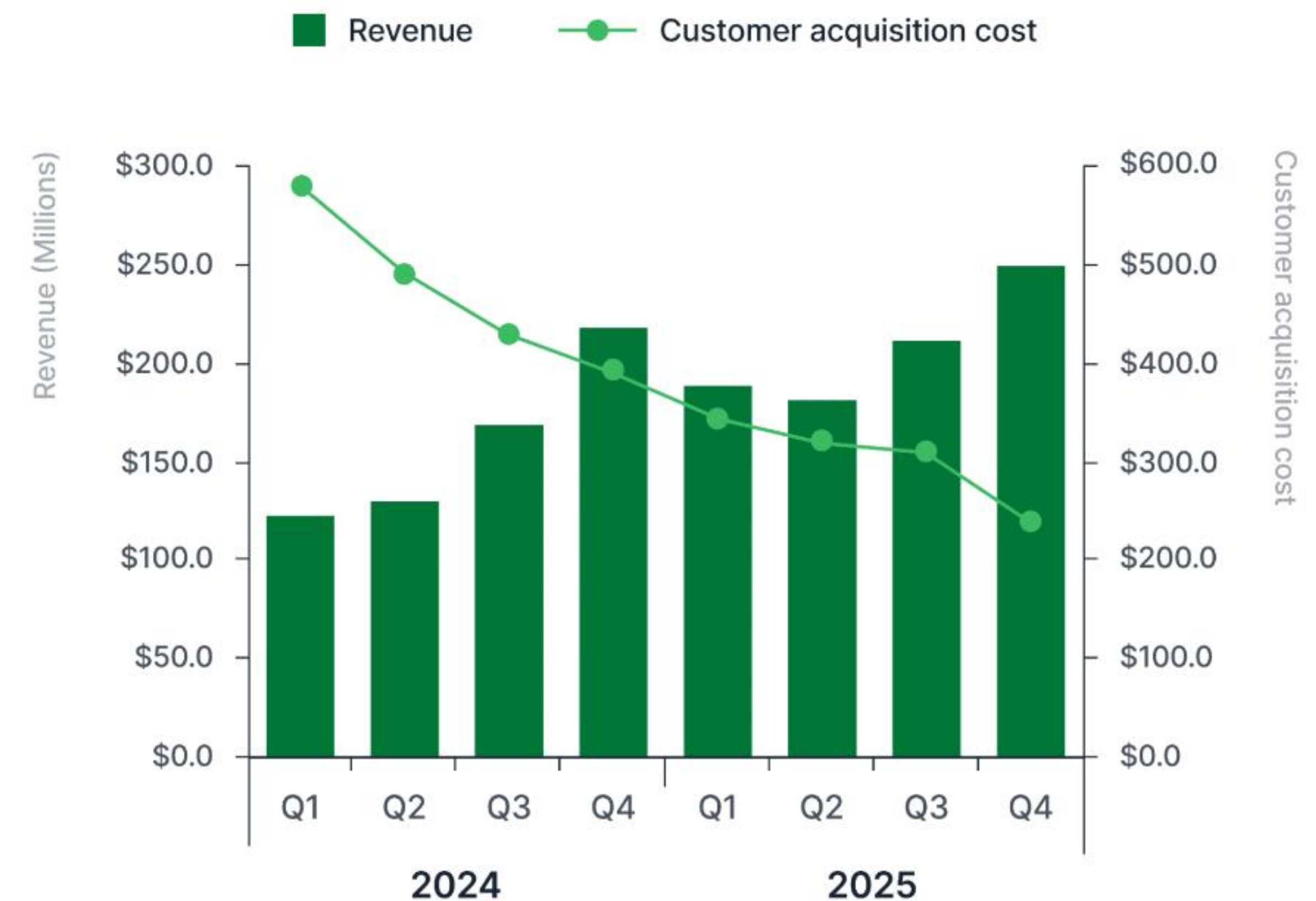
Dataset: `sales_data.csv`

Goal: The VP of Sales needs a performance update.

Tasks:

1. What are the top 5 `Product_Name` by `Sales_Amount` ?
2. What is the `Profit_Margin` for each `Product_Category` ?
3. Who is the top `Sales_Rep` by `Profit` ?
4. Build a dashboard that answers these questions.

Example company:
Revenue vs customer acquisition cost



Beginner Module Complete!

You've Got the Fundamentals!

You now have the fundamental skills to build and share visualizations in Tableau.

Next Steps:

- Practice with your own data!
- Explore the Tableau Public community gallery.
- Get ready for the **Intermediate module**, where we cover Joins, LODs, and more advanced charts.

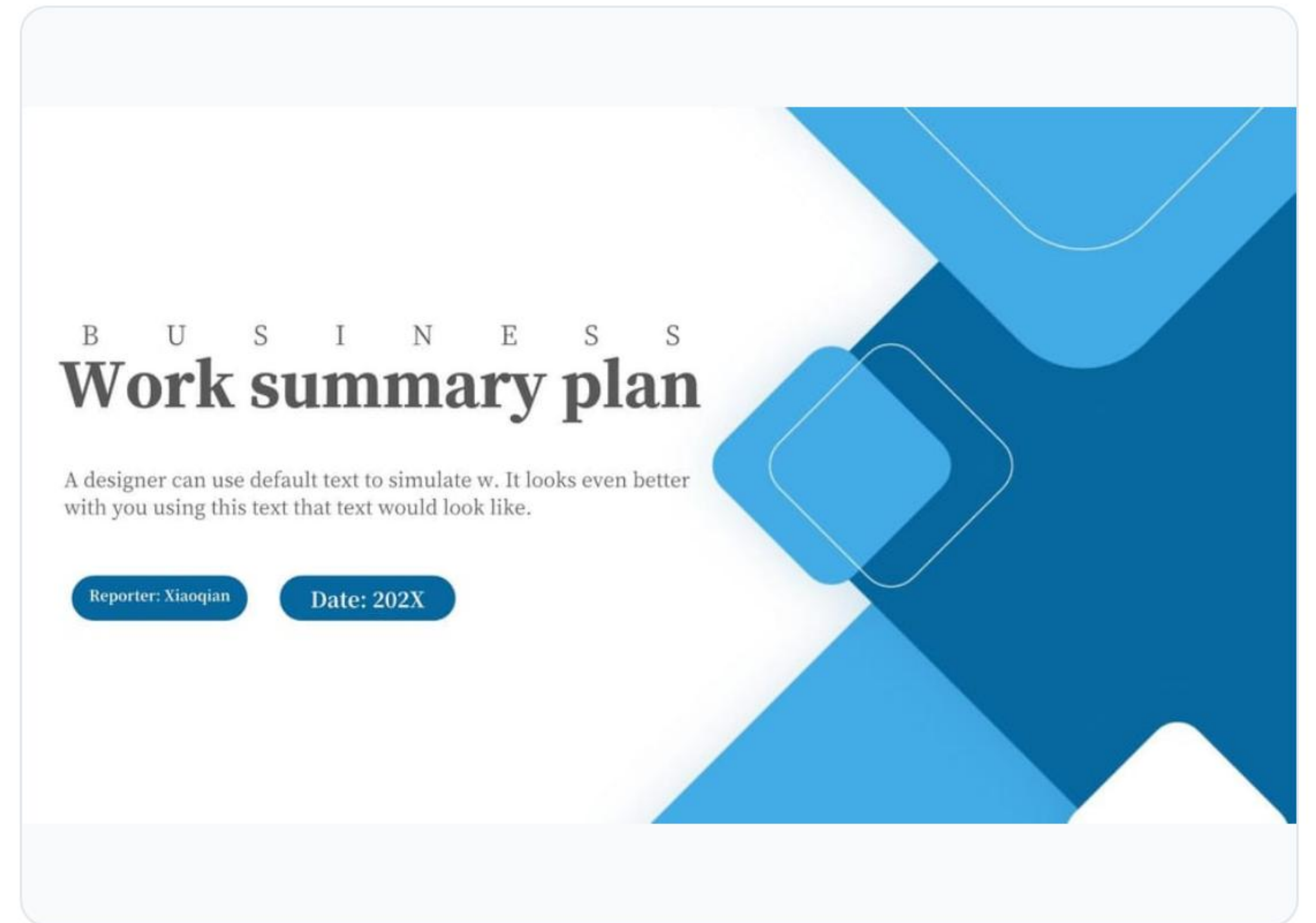


Image Sources



https://png.pngtree.com/png-vector/20250902/ourlarge/pngtree-flat-minimalist-business-growth-icon-with-blue-bar-chart-and-yellow-vector-png-image_17355733.webp

Source: [pngtree.com](https://png.pngtree.com/)



Thumbnail for
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<https://playfairdata.com/wp-content/uploads/2024/10/CPA@2X.png>

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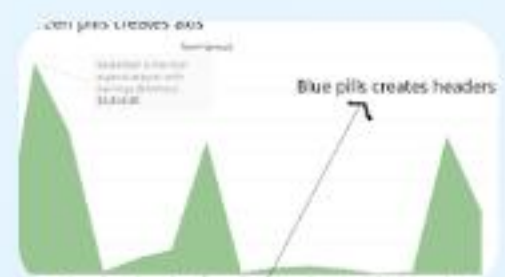
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<https://help.tableau.com/current/pro/desktop/en-us/lmg/showme2.png>

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Image Sources



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https://help.tableau.com/current/pro/desktop/en-us/Img/envIRON_workspace_datasource_page.png

Source: help.tableau.com



<https://user-images.githubusercontent.com/124501309/227755419-ff277f69-fd14-4d51-b85f-973c25a6219f.png>

Source: github.com



Thumbnail for
tableauacademy.substack.com

[https://substackcdn.com/image/fetch/\\$s_!OW-F!,f_auto,q_auto:good,fl_progressive:steep/https%3A%2F%2Fbucketeer-e05bbc84-baa3-437e-9518-adb32be77984.s3.amazonaws.com%2Fpublic%2Fimages%2F7e6fe286-ca87-42e1-b3f3-b5de33b18069_761x572.png](https://substackcdn.com/image/fetch/$s_!OW-F!,f_auto,q_auto:good,fl_progressive:steep/https%3A%2F%2Fbucketeer-e05bbc84-baa3-437e-9518-adb32be77984.s3.amazonaws.com%2Fpublic%2Fimages%2F7e6fe286-ca87-42e1-b3f3-b5de33b18069_761x572.png)

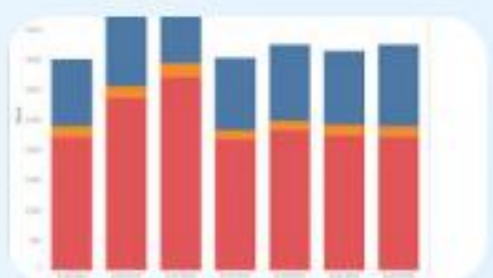
Source: tableauacademy.substack.com



Thumbnail for
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<https://assets.justinmind.com/wp-content/uploads/2025/02/ebay-search-filter-ui-example.png>

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<https://community.tableau.com/sfc/servlet.shepherd/version/renditionDownload?rendition=THUMB720BY480&versionId=0688b00000WQPyY&operationContext=CHATTER&contentId=05T8b00001fQVil&page=0>

Source: community.tableau.com

Image Sources

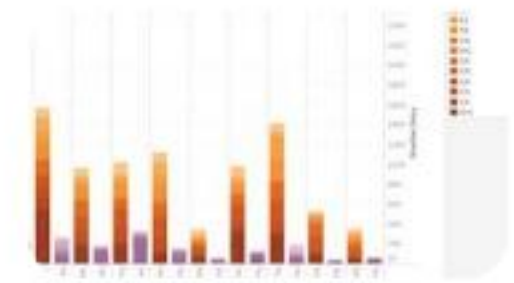


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